



# Sodium chloride (NaCl) as a treatment against trichodiniasis for pike perch (*Sander lucioperca*) in recirculating aquaculture systems

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## Abstract

Pike perch were transferred from an over-wintering net cage into recirculating aquaculture systems (RAS). Within a 21-day period, it was examined whether or not ectoparasites, here specifically a representative of the family of Trichodinidae that were naturally present on the fish, could be reduced or even eliminated from the fish by means of sodium chloride treatment (3 and 6 g NaCl/L). Despite a significant reduction in trichodinid density, the application of 3 g NaCl/L proved to be insufficient to eliminate all trichodinids during the 21-day trial. However, under RAS conditions, a continuous 21-day exposure to 6 g NaCl/L completely eliminated the trichodinids on the fish. Based on the observations of the composition of the trichodinid density, it is recommended to screen fish for ectoparasites on multiple body sites. In conclusion, NaCl can be considered an effective and low-cost therapeutic in pike perch aquaculture against trichodiniasis.

**Keywords** Pike perch · Ectoparasite · Treatment · Trichodiniasis · NaCl · Recirculating aquaculture systems (RAS)

## Introduction

The European pike perch (*Sander lucioperca* (L. 1758)) is a promising candidate for the production in recirculating aquaculture systems (RAS) (FAO 2023). Although RAS are considered closed production systems that have a much lower risk of fish pathogen and parasite outbreaks compared to aquaculture systems using surface waters, invasions of pathogens and parasites can still take place via make-up water or contaminated stocking material (Bebak-Williams and Bowser 2013). Parasites are, in addition to water quality,

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stocking density or feeding regime, an important aspect of fish husbandry in RAS considering animal welfare in aquaculture (Barber 2007). Parasite infestation can have severe impacts on the fish's sensory as well as swimming ability and increase energy demand. As a result of the impairment or as a direct response to the parasite infection, feed intake and competitive strength may be affected. Furthermore, injuries to mucosa, skin, gills or fins caused by parasites are potential entry sites for secondary infections by other fish pathogens (Amlacher 1986; Barber 2007; Hoffmann 2005; Schäperclaus 1990).

In freshwater fish, the application of NaCl can reduce or completely prevent the presence of ectoparasites (Mifsud and Rowland 2008; Schelkle et al. 2011). Also, those ectoparasites specifically associated with pike perch in aquaculture, e.g. *Ichthyophthirius multifiliis*, *Gyrodactylus* sp., *Ichthyobodo necator* or *Trichodina* sp., showed to be vulnerable when exposed to salts (Aihua and Buchmann 2001; Németh et al. 2013; Schäperclaus 1990; Smith and Schwarz 2019; Schelkle et al. 2011). The latter one, trichodinids, are ciliated protozoans (Amlacher 1986) with a body size of up to 100 µm (Hoffmann 2005) and the ability to reproduce asexual by transversal division (Amlacher 1986). When viewed from above, they have a circular body shape and from the side, a bell-shaped appearance (Schäperclaus 1990). Most trichodinids infest the skin and gills of fish by attaching themselves to their host using their sucking disc. However, they do not feed on the host itself, but on the bacteria and organic material present (Collymore et al. 2013). Mass infestations on gills, skin and fins can result in fish loss (Khan 1991; Pearse 1972) which consequently reduces aquaculture production.

For the treatment of trichodiniasis, the application of sodium chloride (NaCl) is recommended (Amlacher 1986; Schäperclaus 1990, Tavares-Dias 2022). However, this procedure often involves either repeated treatments (e.g. 3 g/L; Grignard et al. 1996) or the use of short-term dip baths with relatively high salt concentrations, e.g. 10–15 g NaCl/L for 20–30 min (Schäperclaus 1990). A short-time bath of RAS-reared fish would require additional handling, which subsequently will evoke unnecessary stress for the fish or if improperly applied could even be toxic (Foissner et al. 1985). Furthermore, conducting these baths directly in RAS would require high quantities of salt as well as the need to change a major part of the rearing water to decrease the salt concentration after treatment. Both procedures can be considered economically inefficient. On top of that, salt concentrations above 10 g/L will negatively affect the biofilter performance (Levine and Meade 1976). Despite that the physiological effects of incomplete nitrification on fish can be compensated by salt application (Wuertz et al. 2013), it will still have effects on the rearing environment and production output. Thus, there is a need for an effective and practical protocol of treating trichodinid infestations with low salt concentrations over a prolonged period of time directly in RAS, which in turn could close a knowledge gap for treatment options in pike perch aquaculture (Rożyński et al. 2022).

From the physiological perspective, pike perch are able to tolerate a wide range of salt concentrations as they have significant recovery abilities when exposed to salinity shifts (Demska-Zakeś et al. 2021). For instance, they have shown tolerance for high salinities ranging from 29 to 33 g/L during a simulated tidal cycle of 6 h and for a direct transfer in water with salt concentrations up to 16 g/L for at least 6 days (Brown et al. 2001).

Slight increase of salinity can be considered a good practice when culturing pike perch in RAS (Polcar et al. 2019). Preliminary tests focusing on fish performance indicate that prolonged rearing of pike perch (200–500 g body weight) in RAS using 6 g NaCl/L is possible without any decrease in performance (von Plessen and Müller-Belecke, pers. com.).

This study aimed to determine the effects of NaCl application on pike perch naturally infected with trichodinids as well as their aquaculture performance. It is hypothesized that

applying a continuous NaCl concentration is able to reduce or, ideally, completely eradicate trichodiniasis of pike perch in RAS. Thus, by continuously using low NaCl concentrations to treat these parasites in RAS, both medications and the application of disinfectants can be reduced or eliminated in the future.

## Materials and methods

The experiment and subsequent laboratory work were conducted at the Potsdam Institute of Inland Fisheries (Germany). The research project in which this experiment was conducted and the subsequent animal testing application were registered under the file number 2347–8–2017 at the federal state of Brandenburg (Landesamt für Arbeitsschutz, Verbraucherschutz und Gesundheit).

### Recirculating aquaculture systems

Fish rearing took place in four identical RAS. Each of the systems consisted of three fish tanks (each with a volume of 320 L) and a mechanical/biological treatment unit (120 L of volume) for the removal of particulate matter and nitrification. Fifty liters of biofilter material ( $0.9 \text{ g/cm}^3$ ;  $859 \text{ m}^2/\text{m}^3$ ) was used for each RAS. Prior to the experiment, the biofilter material was acclimatized to a NaCl concentration of 3 g/L in a semi-commercial pike perch RAS for 47 days. Water flow in RAS was maintained using a pump (Eheim Compact 1000) and aeration was applied with a membrane compressor (200 L/min) which was connected to one air stone in each rearing unit ( $D=30 \text{ mm}$ ) and two additional air stones in the biofilter bed ( $D=50 \text{ mm}$ ).

The initial filling of three RAS was done with filtered water from Lake Sacrow (100  $\mu\text{m}$  mesh width, 3 m depth, 16 °C, pH 7.8). In contrast, a fourth RAS named *internal control* was filled with filtered and UV-C conditioned water (254 nm; 42 W;  $1.4\text{--}4.0 \text{ m}^3/\text{h}$ ; Trop Electronic GmbH) from Lake Sacrow. The conditioning step of the water for that RAS took place in a separate tank for 4 days prior to the initial filling.

During the experiment, commercial aqua feed (4.5 mm, crude protein: 49%, crude fat: 10%, crude fibre: 1.5%, ash: 7.9%, phosphorus: 1.27%) was administered according to a protocol by Müller-Belecke and Zienert (pers. com.) using 24-h belt feeders. Daily water exchange was set to 600 L of water/kg of feed administered on the previous day. The exchange of water was done via the mechanical/biological treatment unit. In order to prevent ectoparasites from entering the RAS with the daily make-up water, only filtered and UV-C conditioned water was used during the experiment.

The daily routine work included the following steps: visual inspection of the fish, visual checking for feed and faeces residues, recording of water parameters (temperature °C, oxygen concentration mg/L (WTW Oxi 340i), pH value (Merck MColoritest 5.0–9.0)), checking the pump function, cleaning of the biofilter bed including the manual water exchange, adding of sodium bicarbonate at a pH value  $\leq 7.2$  and administering feed.

In addition to the water parameters that were determined on a daily basis, several other water analyses were carried out at least once per week. The analyses included the determination of the concentrations of ammonium (mg/L, LCK 304/305), nitrite (mg/L, LCK 341) and nitrate (mg/L, LCK 339/340) by means of photometric measurement (Hach, DR5000, LCK cuvette test system). Additionally, acid neutralizing capacity (ANC, by titration of

0.1 mmol/L hydrochloric acid, WTW pH 720), pH value (WTW pH 720) and conductivity ( $\mu\text{S}/\text{cm}$ , WTW LF 197) were determined.

## Experimental fish

In August of the previous year at iso-thermic conditions, RAS-reared pike perch (*Sander lucioperca*) were transferred into a net cage in Lake Sacrow for over-wintering. At the end of May, 57 pike perch were in total transferred into each of the three RAS corresponding to 19 fish per tank. One day after stocking of the fish, iodine-free NaCl (99.8% NaCl incl. E500 and E535; ESCO) was added to the RAS in the following scheme:

RAS I: 0 g NaCl/L (*freshwater control*)

RAS II: 3 g NaCl/L

RAS III: 6 g NaCl/L

RAS IV: 0 g NaCl/L (*internal control*)

The NaCl concentration of RAS III was increased gradually by 3 g/L per day. On the contrary, the RAS called *internal control* was stocked with 57 pike perch individuals that were free of ectoparasites. This status was achieved by having these fish transferred into a separate RAS 7 days before the actual start of the experiment. Subsequently, a therapeutic agent (single application of 10 mL/m<sup>3</sup> formaldehyde solution (37%), CHEMSOLUTE) against ectoparasite was applied.

Fish weight ( $\pm 0.1$  g) and total length ( $\pm 5$  mm) were determined before stocking the fish into RAS, at sampling days where individuals were removed for ectoparasitic analyses and at the end of the experiment. For each tank, the specific growth rate (% BW/d) was calculated as  $\text{SGR} = (\ln(\text{final total biomass}) - \ln(\text{initial total biomass})) / (\text{number of experimental days}) * 100$  and the feed conversion ratio calculated as  $\text{FCR} = (\text{feed administered}) / (\text{weight gain})$ . The condition factor ( $\text{g}/\text{cm}^3$ ) was calculated according to Fulton's formular as  $\text{cf} = (\text{fish weight} * 100) / \text{fish length}^3$  (Nash et al. 2006). Mortality (%) was calculated as  $M = (100 / \text{living individuals}) * (\text{dead individuals})$ .

## Ectoparasite examination

Pike perch were examined for ectoparasites at the start of the experiment (day 0), after 7 and 21 days. The examinations were carried out taking into account common protocols (Amlacher 1986; Hoffmann 2005; Noga 2010; Schäperclaus 1990). On each sampling day, 12 pike perch per RAS (four fish per tank) were examined. In accordance with animal welfare requirements, fish were anaesthetized by a blow on the head and killed by exsanguination (puncturation of the heart and disconnection of the blood circulation) before the examination of the left side of the body. In addition to two organ samples, three mucosal swabs were taken in the direction of the scales by using cover slips (24 × 24 mm):

Back: Swab starting below the 1st dorsal fin to the end of the 2nd dorsal fin

Anal fin: Swab of the skin above the anal fin along its entire width

Ventral fin: Complete swab of the skin area covered by the ventral fin

Pectoral fin: Complete fin

Gill arch: Second gill arch

Using a drop of water, the mucosal swabs were transferred to microscope slides (76×26×1 mm) and smeared out. The organ samples were transferred to microscope slides, wetted with a drop of water and covered with cover slips. The samples were examined using a light microscope (Leitz, Laborlux S; ocular: 10×/20). At a selected magnification of 100×, the ectoparasites were determined and counted on six randomly selected viewing fields. Only viewing fields that were completely covered with mucus or organ samples were considered for the data recording.

Taking into account Bush et al. (1997), the prevalence of trichodinids ( $T_E$ , %) was calculated for each RAS as  $T_E = (\text{number of infected fish} \times 100) / \text{number of fish examined}$ . Since RAS are systems in which tanks are indirectly connected to each other,  $T_E$  was presented for the entire RAS.

In addition, the average density of the ectoparasites was calculated. Following Bush et al. (1997), the average Trichodinidae density per viewing field and fish (TDVF, pc.) was calculated as  $TDVF = ((X_{\text{Fish 1}} + X_{\text{Fish 2}} + \dots + X_{\text{Fish } n}) / \text{number of fish examined})$ . Since a sixfold determination of the number of parasites has taken place in each of the five mucus and organ samples, respectively, a mean value ( $\bar{X}$ ) was calculated for each body site and fish examined.

## Statistics

The experimental setup was chosen to investigate the effects of NaCl concentrations on ectoparasites attached to pike perch.

Statistical analyses and graphical presentations were conducted using SIGMAPLOT 12.0. All data are presented as mean  $\pm$  standard deviation (SD). For statistical analyses, each of the three rearing units of RAS was considered a replicate.

For data with a normal distribution (Shapiro–Wilk) and homogeneous variances (Levene's test), a one-way analysis of variance (ANOVA) was performed. In the presence of significant differences, Tukey's post hoc tests were conducted for a multiple comparison between the groups. In the case of non-normally distributed data or for normally distributed data with heterogeneous variances, a Kruskal–Wallis one-way analysis of variance on ranks was applied. Subsequently, Dunn's post hoc tests were performed for multiple comparisons of the groups. In both cases, the  $p$ -value was set to 0.05.

To test for a correlation between two variables, Spearman's rank-order correlations were performed. The correlation coefficient ( $r$ ) quantifies the strength of the association between the two variables investigated. The  $p$ -value was set to 0.05.

## Results

### Water analyses

The results of the water analyses are displayed in Table 1. Throughout the experiment, there were significant differences in conductivity ( $p \leq 0.001$ ) and oxygen concentration ( $p = 0.003$ ) between the four RAS, but not after the experiment in the fish-free RAS where oxygen concentrations showed no significant differences ( $p = 0.839$ ).

**Table 1** Water parameters of the four RAS. Values are expressed as mean  $\pm$  SD. Values in rows followed by different superscript letters are significantly different

| Parameter                    | 0 g NaCl/L                 | 3 g NaCl/L                  | 6 g NaCl/L                    | <i>Internal control</i>    | Unit       | <i>p</i> -value | <i>n</i> |
|------------------------------|----------------------------|-----------------------------|-------------------------------|----------------------------|------------|-----------------|----------|
| pH                           | 7.3 $\pm$ 0.3              | 7.3 $\pm$ 0.3               | 7.3 $\pm$ 0.2                 | 7.3 $\pm$ 0.2              | -          | 0.969           | 22       |
| Temperature                  | 22.6 $\pm$ 1.5             | 22.6 $\pm$ 1.5              | 22.8 $\pm$ 1.4                | 23.2 $\pm$ 1.1             | °C         | 0.542           | 22       |
| Oxygen <sub>DE</sub>         | 6.9 $\pm$ 0.5 <sup>a</sup> | 7.1 $\pm$ 0.5 <sup>a</sup>  | 7.3 $\pm$ 0.4 <sup>b</sup>    | 7.1 $\pm$ 0.4 <sup>a</sup> | mg/L       | 0.003           | 22       |
| Oxygen <sub>AE</sub>         | 8.2 $\pm$ 0.2              | 8.2 $\pm$ 0.2               | 8.2 $\pm$ 0.1                 | 8.2 $\pm$ 0.1              | mg/L       | 0.839           | 8        |
| ANC                          | 0.77 $\pm$ 0.48            | 1.01 $\pm$ 0.26             | 1.43 $\pm$ 0.41               | 0.86 $\pm$ 0.47            | mmol/L     | 0.168           | 4        |
| NH <sub>4</sub> <sup>+</sup> | 0.39 $\pm$ 0.44            | 0.53 $\pm$ 0.59             | 0.75 $\pm$ 0.71               | 0.29 $\pm$ 0.26            | mg/L       | 0.180           | 7        |
| NO <sub>2</sub> <sup>-</sup> | 0.57 $\pm$ 0.36            | 0.57 $\pm$ 0.30             | 0.71 $\pm$ 0.39               | 0.46 $\pm$ 0.25            | mg/L       | 0.432           | 7        |
| NO <sub>3</sub> <sup>-</sup> | 70.4 $\pm$ 39.7            | 55.4 $\pm$ 38.2             | 46.2 $\pm$ 36.5               | 64.6 $\pm$ 33.8            | mg/L       | 0.503           | 7        |
| Conductivity                 | 456 $\pm$ 33 <sup>a</sup>  | 5636 $\pm$ 148 <sup>b</sup> | 10,415 $\pm$ 219 <sup>c</sup> | 445 $\pm$ 23 <sup>a</sup>  | $\mu$ S/cm | $\leq$ 0.001    | 20       |

Oxygen<sub>DE</sub>, oxygen concentration during the experiment; Oxygen<sub>AE</sub>, oxygen concentration after the experiment in fish-free RAS; ANC, acid neutralization capacity; NH<sub>4</sub><sup>+</sup>, ammonia; NO<sub>2</sub><sup>-</sup>, nitrite; NO<sub>3</sub><sup>-</sup>, nitrate

## Fish performance

The results concerning fish performance are summarized in Table 2. The mean initial body weight varied between 120 and 121 g and the mean final body weight between 125 and 134 g. The specific growth rates ranged from 0.27 to 0.51%/d and the feed conversion ratio from 1.56 to 2.29. No significant differences were found between the four treatments. No mortality was observed during the experiment.

Throughout the experiment, the condition factor (Table 3) of the fish varied between 0.74 and 0.87 g/cm<sup>3</sup>. At the end of the experiment, the fish in 0, 3 and 6 g NaCl/L had a significantly higher condition factor than that of the fish of the *internal control*.

## Trichodinids

A total of 144 pike perch were examined throughout the experiment. During the 21-day trial, no ectoparasites were detected on the fish in the *internal control*.

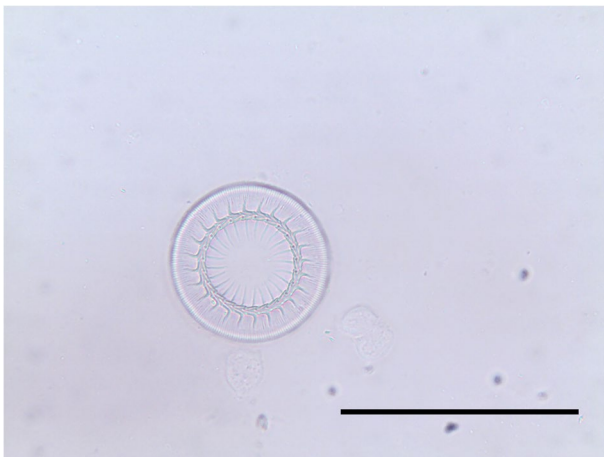
At the start of the experiment, ectoparasites were recorded microscopically on the fish in RAS with NaCl treatment (0, 3 and 6 g/L). Numerous ciliates were detected on the examined fish (Fig. 1). Considering the classification from Lom and Dyková (1992),

**Table 2** Initial and final body weight of individual fish, specific growth rate (SGR, based on tank specific biomass increments), feed conversion ratio (FCR) and mortality of the pike perch in 0, 3 and 6 g NaCl/L and the *internal control*. Values are expressed as mean  $\pm$  SD. There were no significant differences between the groups

| Parameter                      | 0 g NaCl/L      | 3 g NaCl/L      | 6 g NaCl/L      | <i>Internal control</i> | Unit | <i>p</i> -value | <i>n</i> |
|--------------------------------|-----------------|-----------------|-----------------|-------------------------|------|-----------------|----------|
| Body weight <sub>initial</sub> | 120 $\pm$ 16    | 120 $\pm$ 17    | 121 $\pm$ 15    | 121 $\pm$ 14            | g    | 0.869           | 57       |
| Body weight <sub>final</sub>   | 134 $\pm$ 21    | 125 $\pm$ 24    | 132 $\pm$ 24    | 134 $\pm$ 26            | g    | 0.425           | 33       |
| SGR                            | 0.37 $\pm$ 0.05 | 0.27 $\pm$ 0.21 | 0.39 $\pm$ 0.03 | 0.51 $\pm$ 0.05         | %/d  | 0.284           | 3        |
| FCR                            | 2.29 $\pm$ 0.48 | 2.43 $\pm$ 0.39 | 1.95 $\pm$ 0.13 | 1.56 $\pm$ 0.07         | -    | 0.092           | 3        |
| Mortality                      | 0               | 0               | 0               | 0                       | %    | -               | -        |

**Table 3** The condition factor of the pike perch in 0, 3 and 6 g NaCl/L and the *internal control* at different time points during the experiment. Values are expressed as mean  $\pm$  SD. Values in rows followed by different superscript letters are significantly different

| Day | 0 g NaCl/L                   | 3 g NaCl/L                   | 6 g NaCl/L                   | <i>Internal control</i>      | Unit              | <i>p</i> -value | <i>n</i> |
|-----|------------------------------|------------------------------|------------------------------|------------------------------|-------------------|-----------------|----------|
| 0   | 0.74 $\pm$ 0.07              | 0.74 $\pm$ 0.05              | 0.74 $\pm$ 0.06              | 0.76 $\pm$ 0.05              | g/cm <sup>3</sup> | 0.832           | 12       |
| 7   | 0.81 $\pm$ 0.05              | 0.83 $\pm$ 0.04              | 0.82 $\pm$ 0.05              | 0.79 $\pm$ 0.05              | g/cm <sup>3</sup> | 0.163           | 12       |
| 21  | 0.84 $\pm$ 0.05 <sup>a</sup> | 0.85 $\pm$ 0.09 <sup>a</sup> | 0.87 $\pm$ 0.05 <sup>a</sup> | 0.79 $\pm$ 0.06 <sup>b</sup> | g/cm <sup>3</sup> | 0.014           | 12       |

**Fig. 1** Image detail of a representative of the family of Trichodinidae under a light microscope originally taken at 40 $\times$  magnification. Scale bar equals 100  $\mu$ m**Table 4** The prevalence of trichodinids ( $T_E$ , %) on pike perch in four RAS (0, 3, 6 g NaCl/L, *internal control*) at the different time points during the experiment. *n* indicates the number of investigated fish.  $T_E$  values are given for the whole RAS

|                                | Day | 0 g NaCl/L | 3 g NaCl/L | 6 g NaCl/L | <i>Internal control</i> | <i>n</i> |
|--------------------------------|-----|------------|------------|------------|-------------------------|----------|
| Prevalence of trichodinids (%) | 0   | 100        | 100        | 100        | 0                       | 12       |
|                                | 7   | 100        | 100        | 100        | 0                       | 12       |
|                                | 21  | 100        | 16.7       | 0          | 0                       | 12       |

the ciliates were assigned as a member of the family of Trichodinidae. A total of 4565 trichodinids were identified across all treatments and sampling days.

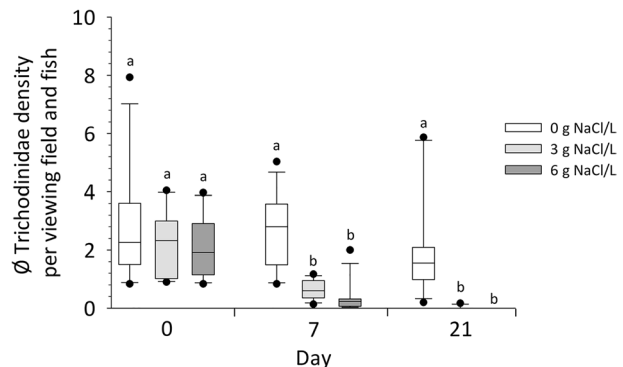
At the start of the experiment, there was no correlation between the fish's condition factor and number of trichodinids detected on the fish ( $r = -0.133$ ;  $p = 0.437$ ;  $n = 36$ ).

Table 4 shows the prevalence of trichodinids per treatment and sampling day. At the beginning of the experiment, all pike perch were infected with trichodinids. As the experiment progressed, the prevalence in the freshwater control (0 g NaCl/L) remained unchanged. The prevalence in the NaCl-exposed groups was unaltered at day 7, but it was clearly reduced 14 days later. While part of the fish in the 3 g NaCl/L treatment continued to be infected after 21 days, no trichodinids were detected on the pike perch in the 6 g NaCl/L treatment.

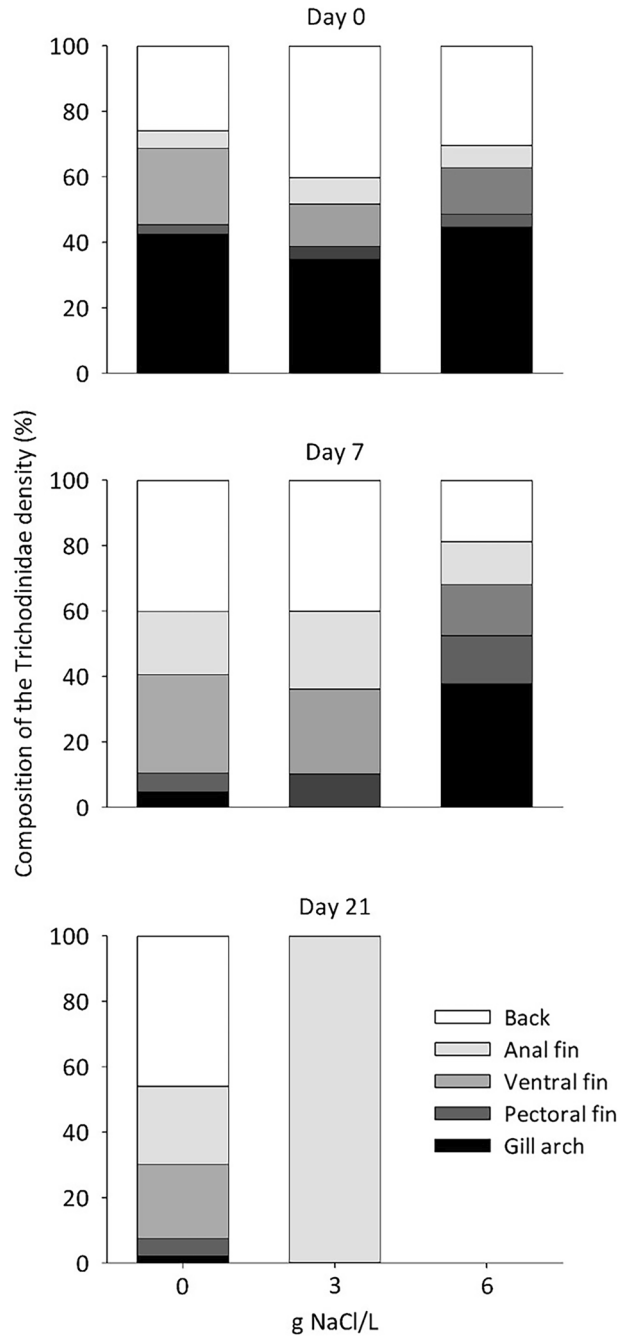
The average Trichodinidae density per viewing field and fish (TDVF) throughout the experiment is displayed in Fig. 2. At the start, the average density of trichodinids was highest in the freshwater control ( $2.9 \pm 1.9$  TDVF). However, no significant differences were found compared to the density in the 3 g NaCl/L ( $2.2 \pm 1.1$  TDVF) and 6 g NaCl/L ( $2.1 \pm 1.0$  TDVF) ( $p=0.648$ ) treatments. The deviation of the TDVF in the freshwater control was the highest as a result of the high number of trichodinids observed on the gills of some fish (up to 60 individuals per viewing field; data not shown separately). Rearing the fish under RAS conditions for 7 days resulted in a 6.4% decrease of TDVF in the freshwater control group ( $2.7 \pm 1.2$  TDVF), a 71.4% decrease in the 3 g NaCl/L ( $0.6 \pm 0.3$  TDVF) treatment and a 83.8% decrease in the 6 g NaCl/L ( $0.3 \pm 0.5$  TDVF) treatment. The TDVF in the 3 and 6 g NaCl/L groups was reduced significantly ( $p \leq 0.001$ ). After 21 days, a TDVF of  $2.0 \pm 1.7$  was determined in the freshwater control group. This value was significantly higher than that in both treatments: 3 g NaCl/L ( $0.02 \pm 0.05$  TDVF) and 6 g NaCl/L (0.0 TDVF) ( $p \leq 0.001$ ). Considering the entire duration of the experiment, there was a reduction of TDVF by 28.6% in the freshwater control, by 99.1% in the 3 g NaCl/L treatment and by 100% in the 6 g NaCl/L treatment. Despite the TDVF decrease by 28.6% in the freshwater control, no significant difference in TDVF on days 0, 7 and 21 was found in this treatment ( $p=0.368$ ).

To visualize the change in TDVF on the different body sites of the fish over time, the composition of this ectoparasite is displayed in Fig. 3. At the start of the experiment, trichodinids were observed on all of the five sampling points that were chosen for investigation. The highest share of densities was found on the gill arch of the fish (35–45%) followed by the back (26–40%), the ventral (13–23%), anal (5–8%) and pectoral fin (3–4%). On day 7 of the experiment, pike perch in the freshwater control and in the RAS with 6 g NaCl/L had trichodinids on all sampling locations but not in the 3 g NaCl/L RAS. In this particular group, no trichodinids were detected on the gill arches (0%). While a reduction down to 5% on the gill was observed in the freshwater control, the gill of the fish in the 6 g NaCl/L group was the body site with the highest share of trichodinids (38%). At the end of the experiment, trichodinids were detected on all five sampling sites in the freshwater group, on one site on the fish (anal fin) in the 3 g NaCl/L treatment. For the 6 g NaCl/L treatment, no trichodinids were detected on any fish.

**Fig. 2** Average Trichodinidae density per viewing field and fish (TDVF) on individual pike perch reared in freshwater RAS with different NaCl treatments at days 0, 7 and 21 of the experiment. 12 fish were examined for ectoparasites per sampling. Significant differences between the treatments at any time point are indicated with a letter



**Fig. 3** Composition of the trichodinid density (%) based on the five sample locations chosen (back, anal fin, ventral fin, pectoral fin, gill arch) on pike perch reared in freshwater RAS with different NaCl treatments at days 0, 7 and 21 of the experiment. 12 fish were examined for trichodinids per sampling



## Discussion

The aim of the study was to investigate the effects of different NaCl applications (0, 3 and 6 g/L) on the prevalence and density of trichodinids of naturally infected pike perch transferred into RAS. No ectoparasites were detected on the fish of the *internal control* at any sampling event. Therefore, it was assumed that there was no cross-contamination between the four RAS, but also no contamination of the systems with trichodinids or other ectoparasites potentially entering with the make-up water. Hence, the experiment can be considered valid.

## Water parameters

The water parameters determined during the experiment were within an acceptable range for this species in RAS (Dalsgaard et al. 2013; FAO 2023). However, significant differences were observed between the systems. Besides the NaCl-induced difference in conductivity, the concentration of dissolved oxygen was also different. Therefore, the four RAS were operated for three additional days after the experiment, during which the oxygen concentration was measured repeatedly. In the fish-free RAS, no differences in oxygen concentrations were detected. Based on this observation, a systematic error can be excluded. It is known that the solubility of oxygen in water depends on its salt concentration and it decreases with an increasing salt concentration. Consequently, the application of NaCl cannot explain the observed differences in oxygen concentrations. However, fish infected with ectoparasites can have an increased oxygen consumption (Grutter et al. 2011; Östlund-Nilsson et al. 2005). For example, Grutter et al. (2011) observed an increase of 35% in oxygen consumption rate in infected fish compared to uninfected individuals. At least two possible explanations for this phenomenon can be mentioned. First, an increase in oxygen demand could be caused by the stress response of fish when infected with ectoparasites (Jørgensen and Buchmann 2007). Second, an ectoparasitic infection could result in an increase of the fish's oxygen need when swimming, as the contact area between fish and the environment is changed (Binning et al. 2013). In addition, the oxygen demand of the experimental fish might have been lower under the effect of the NaCl application, since the energy demand for osmoregulation can be reduced in such environments (Altinok and Grizzle 2003). Thus, the observation of the lowest oxygen concentration in the RAS with the highest density of trichodinids (0 g NaCl/L; freshwater control) as well as the highest oxygen concentration in the RAS with the lowest trichodinid density (6 g NaCl/L) could be explained by the factors mentioned above.

## Fish performance

Concerning fish performance, the *internal control* had the best values for specific growth rate and feed conversion efficiency, but there were no significant differences between the four RAS. The institute's protocol for feeding pike perch in RAS and their subsequent growth performance, indicates a feed conversion ratio of about 1 kg per kg feed administered and a specific growth rate of 1.2 to 1.4% per day for this size class (Müller-Belecke and Zienert, pers. com.). In general, the performance of the pike perch can be considered low.

Growth itself is one of the most important physiological processes in aquaculture. However, intensive production conditions can favour infestations with ectoparasites, which in

turn influence fish performance. For example, fish infected with *Trichodina* sp. had a 26% lower average body weight compared to parasite-free fish at the end of a 12-month trial (Sanmartín Durán et al. 1991). In this experiment, the observed effects of lower growth performance and feed conversion can be attributed to the application of NaCl, as the fish had a necessity to adapt to the new environmental/osmotic conditions (Bœuf and Payan 2001; Brown et al. 2001) and the presence of ectoparasites itself. Additionally, the fish were handled several times during the experiment, which might have had an effect on the performance of the pike perch, in particular as they can be considered to be quite sensitive to handling (Falahatkar et al. 2012). Also, the choice of the fish tanks might not have been ideal for rearing pike perch (narrow tanks with a high water column; low stocking densities which could have favoured the formation of social hierarchies between the fish). Thus, a clear separation of both effects, “NaCl” and “parasite”, on fish performance is not possible.

### Trichodiniasis

In the freshwater control and the RAS with 3 and 6 g NaCl/L treatments, a representative of the family of Trichodinidae was found. These ectoparasites have also been observed on pike perch reared in ponds and natural waters (Movahed et al. 2016; Rahkonen 1994). When infected fish are transferred from open water bodies to RAS, ectoparasites may cause issues in terms of production management and fish health (Németh et al. 2013).

In contrast to results of controlled infestation experiments (Barker et al. 2002), no correlation between the condition factor and the number of trichodinids on the fish was found at the beginning of this experiment. This can be considered an indicator that the ectoparasitic infestation of the pike perch in the over-wintering net cage took place independently of their individual condition.

In the freshwater control, trichodinids were detected on the pike perch at all sampling days, but with a decreasing trend over time. During the 21-day experiment, average trichodinid density decreased by 28.6%. Also, Németh et al. (2013) observed a decrease in *Trichodina* sp. density in the control group (0 g NaCl/L application) during a 14-day experiment with pike perch in RAS. Such a decrease in control groups can be attributed to changing environmental conditions in RAS over time. Water analyses in eel aquaculture facilities, in which the fish were infested with *Trichodina jadranica*, provide evidence that infestations occur more frequently in RAS with higher pH values (6.2–7.4) and lower nitrate concentrations (150–500 mg NO<sub>3</sub><sup>-</sup>/L) compared to facilities with lower pH values (5.4–6.9) and higher nitrate concentrations (250–900 mg NO<sub>3</sub><sup>-</sup>/L) (Madsen et al. 2000). Thus, the decrease of the pH value of the water caused by nitrification and the simultaneous increase of the nitrate concentration could potentially be responsible for the decrease in trichodinid density in the freshwater control (application of 0 g NaCl/L). In the RAS with 3 and 6 g NaCl/L, the environmental parameters, which also changed during the course of the experiment, may have been supportive in addition to the NaCl application.

In freshwater, the application of sodium chloride (NaCl) to the environment is recommended for the control of trichodiniasis. Table 5 gives an overview of the salt concentrations applied and exposure times reported for the trichodiniasis treatment of freshwater fish.

In order to prevent and control infestations in intensive rearing systems with perch (*Perca fluviatilis*), a relative of the pike perch, repeated salt solution baths with 3 g/L are recommended (Grignard et al. 1996). In the present study, a continuous 21-day bath with 3 g NaCl/L was able to significantly reduce the density of ciliates on pike perch (by

**Table 5** Overview of salt concentrations applied (in g/L) and exposure times reported for the treatment of trichodiniasis in freshwater fish

|    | Concentration (g/L) | Specification | Exposure time  | Reference                   |
|----|---------------------|---------------|----------------|-----------------------------|
| 1  | 0.03                | NaCl          | 10 min         | Singhal et al. (1986)       |
| 2  | 10–15               | NaCl          | 20–30 min      | Schäperclaus (1990)         |
| 3  | 10–20               | NaCl          | 20 min         | Amlacher (1986)             |
| 4  | 10–30               | NaCl          | 3–10 min       | García-Magaña et al. (2019) |
| 5  | 10–20               | NaCl          | not specified  | FAO (2023)                  |
| 6  | 30                  | NaCl          | 10 min         | Vargas et al. (2003)        |
| 7  | 3                   | Salt solution | Repeated baths | Grignard et al. (1996)      |
| 8  | 1.5–3.0             | NaCl          | Not specified  | Smith and Schwarz (2019)    |
| 9  | 2                   | NaCl          | 10 h           | Schäperclaus (1990)         |
| 10 | 5                   | NaCl          | 14 days        | Németh et al. (2013)        |
| 11 | 6                   | NaCl          | 21 days        | This study                  |

99.1%), but a complete reduction of the parasites to zero was not possible. It is likely that raising and maintaining the NaCl concentration at 3 g/L are insufficient to completely stop the prevalence of trichodinids. Even though the results of the 3 and 6 g NaCl/L exposure groups were not significantly different at the end of the trial, re-infestations may occur even at a very low prevalence (Jiménez-García et al. 2012), especially since trichodinids reproduce asexually. Whether or not repeated exposures to 3 g NaCl/L (cf. Grignard et al. 1996) would be sufficient to control trichodinids on pike perch in RAS needs to be further investigated in the future. Though, results from the 6 g NaCl/L group indicate that a higher NaCl concentration is needed to treat trichodiniasis effectively in RAS.

In the 6 g NaCl/L RAS, a significant decrease in trichodinids was initially observed on day 7 and a complete reduction by day 21. Furthermore, Németh et al. (2013) described a complete eradication of *Trichodina* sp. on juvenile pike perch treated in RAS using NaCl, but at a lower concentration (5 g/L) and in a shorter time period (14 days). However, to prevent cross-contamination between the fish rearing units of the RAS, the experimental setup included two 18-W UV-C lamps connected in series. For this reason, the observed effect of full eradication of trichodinids by Németh et al. (2013) cannot be attributed exclusively to the permanent NaCl application in the RAS. In particular, another experiment has shown that a transmission of trichodinids, i.e. *Trichodina truttae*, from infected to uninfected salmonids (*Oncorhynchus keta*) can completely be prevented using UV-C radiation (Mizuno et al. 2019).

In addition, one important but yet a subjective observation during sampling events was that the fish in the 6 g NaCl/L RAS had a lesser mucosal thickness than their conspecifics in the freshwater control and 3 g NaCl/L RAS. Rather, the fish of this treatment felt “dry” when handled for analyses. Schäperclaus (1990) has also described this phenomenon for fish treated in hypertonic salt solutions. As a result of the elevated salt concentrations, fish skin slimes heavily, causing ectoparasites to be sloughed together with the mucus. However, the parasites will still be in the RAS and could possibly infect other fishes (Mizuno et al. 2019); it is reasonable to assume that an application of 6 g NaCl/L into RAS is able to treat an infestation with trichodinids by freeing the fish from this parasite while additionally providing an effective reduction of the parasites through an osmotic pressure difference (Buchmann 2022).

In addition to the density of trichodinids, the composition of the ectoparasite density on the fish bodies was investigated. Besides the reduction of density on the fish in the fresh-water control, trichodinids were present on all sampling locations and at all the sampling events, yet with different distributions. For example, the density of trichodinids attached to the gill decreased whereas the density on the back increased throughout the experiment. Also, for the fish exposed to NaCl, a change in the composition of the ectoparasite density was observed, but not associated with a clear pattern. In this context, it is speculated that the application of 0, 3 and 6 g NaCl/L might have resulted in a dose-dependent change of the mucosal characteristics, which in turn might have affected the parasite density of the treatment groups differently. Irrespectively of that, the results indicate that it is recommended to always check for ectoparasite infestations on several body sites of fish.

## Conclusion

Compared with the *internal control*, growth and feed conversion of pike perch infected with ectoparasites were insignificantly lower in the RAS with NaCl application. It is hypothesized that both the parasite infestation and the necessity to adapt to new osmotic conditions were responsible for this observation. From the pike perch infected with ectoparasites, fish treated with 6 g NaCl/L had the best feed conversion and the highest specific growth rates.

In the freshwater control, the density of the trichodinids decreased over time even without the application of NaCl. This decrease in density is probably caused by the changing environmental conditions in RAS over time. When applying 3 g NaCl/L for 21 days in RAS, the trichodinid density was significantly reduced but not completely eliminated; thus, subsequently re-infestations cannot be excluded. Hence, a continuous exposure with 3 g NaCl/L for 21 days is not sufficient to eliminate trichodiniasis of pike perch in RAS. However, this experiment has demonstrated that a continuous application of 6 g NaCl/L removed all trichodinids on pike perch with a body weight of 120 g reared under RAS conditions without the use of any other chemicals and drugs. Since no further sampling had taken place between days 7 and 21, it can be precautionarily assumed that a continuous exposure with 6 g NaCl/L for 21 days at water temperatures of 22–23 °C must be applied for a successful treatment of trichodiniasis in RAS-cultured pike perch. Based on the results of the ectoparasitic composition, it is recommended for aquaculture practice to screen for trichodinids on multiple body sites on fish. In summary, NaCl can be considered an effective and low-cost alternative with the potential to replace conventional therapeutics in pike perch aquaculture.

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**Data availability** The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval** The research project in which this experiment was conducted and the subsequent animal testing application, were registered under the file number 2347–8-2017 at the federal state of Brandenburg (Germany).

**Competing interests** The authors declare no competing interests.

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